Yash Bavadiya

+91-9879163975 krbavadiya11@gmail.com Portfolio LinkedIn GitHub X LeetCode

EDUCATION

Indian Institute of Technology, Jodhpur **Aug 2024 – Jul 2028**
Bachelor of Technology - Computer Science and Engineering - **CGPA: 7.21** Jodhpur, Rajasthan

TECHNICAL SKILLS

Languages: C++, Rust, TypeScript, JavaScript, Python, C, SQL
Systems & AI: Distributed Systems, Discrete-Event Simulation, Concurrency, OS, Networks, DBMS, RAG, LLM Agents, MCP
Technologies: Qt, React, Node.js, WebAssembly, PostgreSQL, SQLite, Redis, WebSockets, Docker, Linux, GitHub Actions

EXPERIENCE

Workers IO (YC F26) **Sep 2026 – Present**
Developer Relations Engineer | Deterministic simulation testing Remote (San Francisco)

- Built **Breakscale**, an open-source system design simulator that led to a founder-led hire, and scaled it to **2,000+ GitHub stars** and 168 forks with **1,200+ tests** and 31 PRs from **15 outside contributors**.
- Engineered a **discrete-event engine** modeling queue buildup, timeouts, retry storms, and circuit-breaker trips dynamically across **33 component types**.
- Shipped **fault injection** (crash, slow, partition, packet loss) with **deterministic replay links**, allowing the exact failure to be replayed from a single shareable URL using identical seeds and fault logs.
- Launched **breakscale-mcp** (an **MCP server** for AI code simulation) and a **VS Code extension**, alongside Breakscale's acceptance into the Vercel, Claude, and Mintlify open-source programs.

Google Summer of Code 2026 **May 2026 – Aug 2026**
Kdenlive @KDE Community | Open Source Contributor, C++ and Qt Remote

- Designed and shipped **per-channel RGB curves**, a **multi-stop gradient editor**, and **time-remap keyframe types** in **C++ and Qt**, wired into Kdenlive's MLT filter pipeline for **Kdenlive 26.08** (41,000 monthly Flathub installs).
- Resolved an **11-year-old bug** via an interval-merging **gap-detection algorithm**, landing **23 merge requests** through close collaboration and code review with core maintainers.

Undergraduate Researcher **Jul 2025 – Nov 2025**
Indian Institute of Technology, Jodhpur | Supervisor: Dr. Pratik Mazumder Jodhpur

- Developed the retrieval pipeline of an **offline RAG system** over 300-page PDFs using **FAISS** vector search and a **4-bit quantized Phi-3**, achieving ∼**3 s** latency on a laptop GPU and a **+71.77%** ROUGE gain over baseline.

PROJECTS

HatchZip | GitHub | Website | **Rust, Slint, ZIP/7z/RAR/tar** **Oct 2026**

- Architected an archive extractor that **frees disk space during extraction** by saving the index aside, extracting backward, and **truncating the archive** after each batch, or **punching holes** in `.tar.gz` streams.
- Guaranteed **crash safety** via ordered writes and rigorous disruption testing, implementing **130+ tests** to verify freed space across ext4, APFS, and NTFS filesystems.
- Achieved **5x faster** extraction than `unzip`, reaching **2,700+ stars**, **27,000+ downloads**, and the **#1 new Rust repo** ranking while patching **2 security advisories**.

gyotaku | GitHub | Website | **Rust, GPUI, ONNX Runtime, SQLite FTS5** **Oct 2026**

- Engineered offline search over screenshots by processing text through **PaddleOCR on ONNX Runtime** into a **SQLite FTS5 trigram index** that highlights matches in place.
- Optimized search latency to **1–10 ms per keystroke** across 5,000+ screenshots at **37 MB idle memory**, shipping across macOS, Windows, and Linux to achieve **390+ stars** and **4,000+ downloads**.

Flock | GitHub | npm | **TypeScript, WebSocket, Redis, React, Docker** **Jun 2026**

- Implemented multiplayer cursors and presence in a **10-line React integration**, scaling servers horizontally via a **Redis pub/sub relay** using **keyspace notifications** to evict dropped clients.
- Delivered throttling, mid-animation cursor smoothing, and **exponential-backoff reconnects**, publishing 3 npm packages to achieve **1,100+ downloads** alongside **58 tests**.

Game Boy Emulator | GitHub | **C++20, SDL2, CMake** **Jul 2026**

- Emulated the Game Boy from scratch featuring a CPU with all **501 opcodes**, 5 interrupts, a DIV-based timer, PPU, and **MBC1/MBC3** banking, successfully passing all **11 Blargg CPU tests** and booting Pokémon Red.

ACHIEVEMENTS

- **Open source:** Contributed merged fixes to Deno, Adobe React Spectrum, Kubernetes SIGs, nlohmann/json, Vitest, Nuxt, and jj.
- **Building in public:** Grew to **3,000+ X followers** and **3.9M impressions** in 3 months, including a **1.9M-view** launch.
- Ranked AIR **3449** of 180,000+ in JEE Advanced 2024 and **99.65** percentile in JEE Main, while solving **450+** LeetCode problems and reaching a **1429** Codeforces rating.
- Mentored **30 students** in open source as a Junior Core Team Member at DevlUp Labs, serving additionally as a Core Member for both the Programming Club and E-Cell.